Gray (1996) and Klaassen (2002)

Collapsing the regime path: models, derivations and results

A reading summary for the MS-GARCH thesis · 13 October 2026

Gray, S. F. (1996). Modeling the conditional distribution of interest rates as a regime-switching process. *Journal of Financial Economics*, 42, 27–62.
Klaassen, F. (2002). Improving GARCH volatility forecasts with regime-switching GARCH. *Empirical Economics*, 27, 363–394.

The two papers in one paragraph

Both papers want GARCH dynamics inside each regime of a Markov-switching model, and both run into path dependence: with a lagged-variance term, the regime variance at t depends on the whole unobserved regime history. **Gray** breaks the dependence by replacing the lagged regime variance with the *aggregate* variance of the previous observation, averaged over regimes. The likelihood then has a first-order recursive structure, like ordinary GARCH. On weekly US T-bill rates, his model splits the data into a high-volatility regime with fast-decaying shocks and a low-volatility regime with slower ones; single-regime GARCH persistence above one becomes 0.4–0.7 within regimes. **Klaassen** keeps Gray's idea but averages more cleverly: he waits until $t-1$ to integrate out the previous regime and conditions on the *current* one. That raises the likelihood and, above all, makes multi-step variance forecasts a simple recursion. On daily dollar exchange rates he shows that GARCH forecasts are too high in volatile periods, and that the regime-switching GARCH removes this bias and forecasts better out of sample.

How to read this summary.

- **Notation.** Equations are in thesis notation; translation tables at the end map them to each paper's symbols.
- **Regime numbering. Gray's regime 1 is the high-volatility regime; Klaassen's regime 1 is the low-variance regime.** I keep each paper's own numbering so the tables match the PDFs.
- **Sources of numbers.** Numbers are taken from the papers and were checked against the page images, because the text layer of both PDFs drops minus signs. Figures I computed from the published estimates are marked *(computed)*.
- **My own material.** Derivations and interpretations that are mine rather than the authors' are in orange boxes.

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1 The common starting point: path dependence

Take a two-regime model in which every GARCH parameter switches:

$$\varepsilon_t = \sigma_t(S_{1:t})\,\eta_t, \qquad \sigma_t^2(S_{1:t}) = \omega_{S_t} + \alpha_{S_t}\varepsilon_{t-1}^2 + \beta_{S_t}\,\sigma_{t-1}^2(S_{1:t-1}). \tag{1}$$

The lagged variance $\sigma_{t-1}^2(S_{1:t-1})$ was itself computed with the regime at $t-1$ and the variance at $t-2$, and so on back to the start. Gray draws this as a tree (his Fig. 1): from a starting variance, two branches lead to $t=1$; each of these splits again at $t=2$; and the branches never rejoin, because the variance on each depends on which regime it came from. With K regimes there are K^t distinct variances at date t, and the likelihood contribution

$$f(\varepsilon_t \mid \mathcal{F}_{t-1}) = \sum_{S_{1:t}} f(\varepsilon_t \mid S_{1:t}, \mathcal{F}_{t-1})\, P(S_{1:t} \mid \mathcal{F}_{t-1})$$

has K^t terms. Cai (1994) and Hamilton & Susmel (1994) avoided this by dropping the GARCH term. Gray and Klaassen keep it and instead **make the tree recombine**: at each date they replace the path-dependent quantity with something that depends only on the current regime, so that only K variances exist at each date (Gray's Fig. 2).

The two papers differ in exactly *what* they replace $\sigma_{t-1}^2(S_{1:t-1})$ with. That difference drives everything else: the likelihood, forecasting, fit and the critiques.

Part I
Gray (1996)

2 The question

Gray models the weekly change in the one-month US Treasury bill rate. Two stylised facts must be captured: the short rate mean-reverts, and changes in it are strongly leptokurtic (sample kurtosis 25.4). Existing models handle the second fact through conditional heteroskedasticity, either via the *level* of the rate (diffusion models such as Cox–Ingersoll–Ross, CIR) or via lagged shocks (GARCH). When fitted to short rates, GARCH estimates often imply $\alpha + \beta \geq 1$. Gray cites values of 1.0096, 1.073 and 1.10 from earlier studies.

His argument, citing Lamoureux & Lastrapes (1990, *JBES*), is that this near-explosive persistence may be an artifact: a shift in the variance process (e.g. the 1979–82 Federal Reserve experiment) is mistaken for volatility clustering. He goes further than a shift in the unconditional variance. All GARCH parameters may differ by regime: the ARCH coefficient could be driven up by short, dramatic, low-persistence episodes while the GARCH coefficient stays high because of persistence in quieter periods. The "generalized regime-switching" (GRS) model is built to let the data show this.

3 The model

3.1 Mixture structure

Write R_t for the short-rate level (annualised %) and $y_t = \Delta R_t$ for its weekly change. The regime $S_t \in \{1,2\}$ is unobserved and $\mathcal{F}_{t-1} = \{R_{t-1}, R_{t-2}, \dots\}$ contains no regimes. Conditional on regime k the change is normal, so conditional on observables alone it is a two-component mixture:

$$y_t \mid \mathcal{F}_{t-1} \;\sim\; \begin{cases} N(\mu_{1,t},\, \sigma_{1,t}^2) & \text{with probability } \xi_{1,t|t-1}, \\ N(\mu_{2,t},\, \sigma_{2,t}^2) & \text{with probability } 1-\xi_{1,t|t-1}, \end{cases} \tag{2}$$

where $\xi_{1,t|t-1} = P(S_t = 1 \mid \mathcal{F}_{t-1})$ is the *ex ante* regime probability.

3.2 Conditional means

Each regime has its own mean reversion:

$$\mu_{k,t} = \lambda_{0k} + \lambda_{1k} R_{t-1}. \tag{3}$$

Mean reversion requires $\lambda_{1k} < 0$. The implied long-run mean of regime k is the level at which the expected change is zero, $-\lambda_{0k}/\lambda_{1k}$.

3.3 Conditional variances: the collapse

Step 1: the variance of the mixture. By the law of total variance,

$$\sigma_t^2 \equiv \mathrm{Var}(y_t \mid \mathcal{F}_{t-1}) = \underbrace{E\big[\mathrm{Var}(y_t \mid S_t, \mathcal{F}_{t-1}) \mid \mathcal{F}_{t-1}\big]}_{\text{average within-regime variance}} + \underbrace{\mathrm{Var}\big(E[y_t \mid S_t, \mathcal{F}_{t-1}] \mid \mathcal{F}_{t-1}\big)}_{\text{dispersion of regime means}}.$$

Evaluating both terms gives Gray's eq. (8):

$$\sigma_t^2 = \xi_{1,t|t-1}\big(\mu_{1,t}^2 + \sigma_{1,t}^2\big) + (1 - \xi_{1,t|t-1})\big(\mu_{2,t}^2 + \sigma_{2,t}^2\big) - \big[\xi_{1,t|t-1}\mu_{1,t} + (1 - \xi_{1,t|t-1})\mu_{2,t}\big]^2. \tag{4}$$

A more transparent form (my algebra)

Writing $\xi_k = \xi_{k,t|t-1}$, the last two terms of (4) simplify for two regimes, because $\xi_1\mu_1^2 + \xi_2\mu_2^2 - (\xi_1\mu_1 + \xi_2\mu_2)^2 = \xi_1\xi_2(\mu_1 - \mu_2)^2$. Hence

$$\sigma_t^2 = \xi_1\sigma_{1,t}^2 + \xi_2\sigma_{2,t}^2 + \xi_1\xi_2\,(\mu_{1,t} - \mu_{2,t})^2.$$

The aggregate variance is the probability-weighted regime variance plus a penalty for uncertainty about which mean applies. When the regime means are equal, as in Klaassen and in the thesis's zero-mean returns, the penalty vanishes and σ_t^2 is just $\sum_k \xi_{k,t|t-1}\sigma_{k,t}^2$.

Step 2: use the aggregate as the lagged variance. The key observation is that σ_t^2 in (4) does *not* depend on the regime path. It is a function of observables and of the two current regime variances only. Gray therefore uses it in place of the path-dependent lagged variance:

$$\boxed{\sigma_{k,t}^2 = \omega_k + \alpha_k\,\varepsilon_{t-1}^2 + \beta_k\,\sigma_{t-1}^2, \qquad k = 1, 2,} \tag{5}$$

where σ_{t-1}^2 is (4) dated $t-1$ (built with $\xi_{1,t-1|t-2}$) and the lagged shock is measured from the *mixture* mean:

$$\varepsilon_{t-1} = y_{t-1} - E[y_{t-1} \mid \mathcal{F}_{t-2}] = y_{t-1} - \big[\xi_{1,t-1|t-2}\mu_{1,t-1} + (1 - \xi_{1,t-1|t-2})\mu_{2,t-1}\big]. \tag{6}$$

Both inputs on the right of (5) are regime-free, so the regime-k variance depends only on the current regime k. Only two variances exist at each date, and the tree recombines.

Read this carefully: what exactly is collapsed

- Two things are aggregated in Gray's model: the lagged *variance* (σ_{t-1}^2, averaged over regimes with ex ante weights $\xi_{\cdot,t-1|t-2}$) and the lagged *shock* (ε_{t-1}, centred on the mixture mean).
- The weights use information up to $t-2$ only. The observation y_{t-1} itself, which is informative about which regime generated it, is not used to decide how to average the variances that produced it. This is the first of the two points Klaassen changes.

- The weights also ignore the regime one is currently computing ($S_t = k$). This is the second point Klaassen changes.

3.4 Level effects (the full GRS model)

To nest the CIR model, the most general regime variance adds the lagged level of the rate:

$$\sigma_{k,t}^2 = \omega_k + \alpha_k \varepsilon_{t-1}^2 + \beta_k \sigma_{t-1}^2 + \psi_k^2 R_{t-1}. \tag{7}$$

The elasticity of the diffusion model is fixed at 0.5 (the CIR value) rather than estimated, because it is almost perfectly collinear with the level coefficient (estimated correlations 0.98–0.99). In the estimated level-effect models ω_k is dropped: it always went to the boundary of zero, since $\psi_k^2 \min(R_{t-1})$ already acts as an intercept.

A discrepancy to be aware of.

- Gray's eq. (9) prints the level term as $\sigma_i^2 \sqrt{r_{t-1}}$.
- The model notes under Tables 4 and 5 write it as $\sigma_i^2 r_{t-1}$, and so does the text, which says that at a 5% rate the regime-2 term equals 0.009. With the reported coefficient 0.0420, $0.0420^2 \times 5 = 0.0088$, which matches $\sigma^2 r_{t-1}$, not $\sigma^2 \sqrt{r_{t-1}}$ *(computed).*
- The CIR model also has variance proportional to r.

So the estimated model uses $\psi_k^2 R_{t-1}$, and Tables 4–5 report ψ_k (not ψ_k^2).

3.5 Transition probabilities

With constant probabilities, $p_{11} = P$ and $p_{22} = Q$ in Gray's notation. In the full GRS model they depend on the level of the rate through the standard normal cdf Φ, which keeps them in $(0, 1)$:

$$p_{11,t} = \Phi(c_1 + d_1 R_{t-1}), \qquad p_{22,t} = \Phi(c_2 + d_2 R_{t-1}). \tag{8}$$

Regime switching can itself create mean reversion. Suppose each regime has a constant mean μ_k. Then

$$\begin{aligned} \operatorname{cov}(y_t, R_{t-1}) &= \operatorname{cov}\big(E[y_t \mid \mathcal{F}_{t-1}], R_{t-1}\big) = \operatorname{cov}\big(\xi_{1,t|t-1}\mu_1 + (1 - \xi_{1,t|t-1})\mu_2,\ R_{t-1}\big) \\ &= (\mu_1 - \mu_2) \operatorname{cov}(\xi_{1,t|t-1}, R_{t-1}). \end{aligned}$$

If the regime probability co-moves with the level, switching produces apparent mean reversion ($\operatorname{cov} < 0$) without any $\lambda_{1k} < 0$. Gray notes that the same argument applies to apparent conditional heteroskedasticity, which is the thesis's theme in the variance.

4 Estimation

4.1 The ex ante probability recursion

All regime information in the past is summarised in one number, $\xi_{1,t|t-1}$, which is updated recursively. Write $f_{k,t} = f(y_t \mid S_t = k, \mathcal{F}_{t-1})$ for the normal density of regime k.

Prediction. By the Markov property,

$$\xi_{1,t|t-1} = \sum_j P(S_t = 1 \mid S_{t-1} = j)\, \xi_{j,t-1|t-1} = p_{11}\, \xi_{1,t-1|t-1} + (1 - p_{22})\,(1 - \xi_{1,t-1|t-1}).$$

Update (Bayes). The filtered probability uses last period's observation:

$$\xi_{1,t-1|t-1} = \frac{f_{1,t-1}\,\xi_{1,t-1|t-2}}{f_{1,t-1}\,\xi_{1,t-1|t-2} + f_{2,t-1}\,(1-\xi_{1,t-1|t-2})}.$$

Substituting the update into the prediction gives Gray's eq. (11) (and (A.6)):

$$\begin{aligned}\xi_{1,t|t-1} = (1-p_{22,t})\left[\frac{f_{2,t-1}(1-\xi_{1,t-1|t-2})}{f_{1,t-1}\xi_{1,t-1|t-2} + f_{2,t-1}(1-\xi_{1,t-1|t-2})}\right] \\ + p_{11,t}\left[\frac{f_{1,t-1}\xi_{1,t-1|t-2}}{f_{1,t-1}\xi_{1,t-1|t-2} + f_{2,t-1}(1-\xi_{1,t-1|t-2})}\right].\end{aligned} \tag{9}$$

The two bracketed terms are the filtered probabilities $P(S_{t-1}=2 \mid \mathcal{F}_{t-1})$ and $P(S_{t-1}=1 \mid \mathcal{F}_{t-1})$. Time-varying transition probabilities slot in without changing the structure, because the regime process is still first-order Markov.

4.2 The likelihood

The log-likelihood is a sum of mixture densities (eq. A.7):

$$\ell_T(\theta) = \sum_{t=1}^{T} \ln\left[\xi_{1,t|t-1}\,\frac{1}{\sqrt{2\pi\sigma_{1,t}^2}}e^{-(y_t-\mu_{1,t})^2/(2\sigma_{1,t}^2)} + (1-\xi_{1,t|t-1})\,\frac{1}{\sqrt{2\pi\sigma_{2,t}^2}}e^{-(y_t-\mu_{2,t})^2/(2\sigma_{2,t}^2)}\right]. \tag{10}$$

Gray's filter, one period at a time

Carry forward $\xi_{1,t-1|t-2}$, $\mu_{k,t-1}$, $\sigma^2_{k,t-1}$ and y_{t-1}.

1. Aggregate last period: σ^2_{t-1} from (4) and ε_{t-1} from (6), both with weights $\xi_{\cdot,t-1|t-2}$.
2. New regime variances: $\sigma^2_{k,t}$ from (5) or (7); new means $\mu_{k,t}$ from (3).
3. New ex ante probability $\xi_{1,t|t-1}$ from (9), using $f_{k,t-1}$.
4. Evaluate $f_{k,t}$ at y_t and add the log of the mixture to ℓ_T.

- **Start-up.** The initial probability can be estimated as a parameter or set to the unconditional probability.
- **Optimisation.** Gray uses BFGS (checked against BHHH and different starting values) with White heteroskedasticity-consistent standard errors.
- **Asymptotics.** For consistency and asymptotic normality of the quasi-ML estimator he cites his 1995 working paper, which is not in the folder.
- **GARCH analogy.** Gray stresses that $\xi_{1,t|t-1}$ plays the same role as σ_t^2 in GARCH: an unobserved parameter of the conditional distribution that follows a known recursion.

4.3 Three kinds of regime probability

Gray distinguishes the *ex ante* probability $\xi_{1,t|t-1}$ (used in forecasting), the *filtered* probability $\xi_{1,t|t}$ (common in the literature) and the *smoothed* probability $\xi_{1,t|T}$ (for dating regimes). He computes smoothed probabilities with a forward-looking algorithm from his 1995 paper; Kim's (1994) backward recursion gives the same result.

5 Results

5.1 Data

Wednesday noon one-month T-bill yields from the Federal Reserve/CRB database, January 1970–April 1994 (1,267 weekly observations). Weekly changes have mean -0.0034, variance 0.1221, skewness -1.51, kurtosis 25.38 and $\text{corr}(y_t, R_{t-1}) = -0.058$. Volatility is visibly high in 1973–75 (OPEC), 1979–82 (Fed experiment) and after October 1987.

The analysis builds up in three steps: constant variance within regimes, then GARCH within regimes, then GARCH plus level effects and time-varying transition probabilities. Each step is compared with its single-regime counterpart.

5.2 Step 1: constant variance within regimes (Table 2)

	Single regime		Regime switching	
Parameter	Est.	t	Est.	t
λ_{01}	0.0475	1.18	0.1687	1.21
λ_{02}	—		-0.0057	-0.31
λ_{11}	-0.0072	-1.07	-0.0190	-1.28
λ_{12}	—		0.0015	0.47
std. dev. regime 1	0.3489	13.74	0.6716	8.58
std. dev. regime 2	—		0.1496	19.68
p_{11}			0.9680	53.48
p_{22}			0.9905	202.14
log-likelihood	-463.14		111.11	
Ljung–Box $Q^2(15)$ $[p]$	221.39	[0.000]	13.44	[0.569]

t-statistics for p_{11}, p_{22} test against 0.5. The two "std. dev." rows are Gray's b_{01}, b_{02}; see the note below.

Reading the variance rows

Gray's notation $N(\cdot, b_{0i})$ suggests that b_{0i} is a variance, but the numbers only make sense as *standard deviations*. The single-regime value 0.3489 equals $\sqrt{0.1221}$, the sample standard deviation. The text says that the regime-1 standard deviation is "more than four times" that of regime 2, and indeed $0.6716/0.1496 = 4.49$; in variances the ratio would be about 20 *(computed)*. I read this as a labelling inconsistency in the paper.

Interpretation.

- **Regime 1: high volatility and high rates.** The long-run mean is $-\lambda_{01}/\lambda_{11} = 8.88\%$ *(computed)*; Gray reports 8.89%. There is weak mean reversion: the expected weekly change is -5.93 basis points at a 12% rate and -11.63 at 15%.
- **Regime 2: low volatility and low rates.** $\lambda_{12} > 0$, i.e. *negative* mean reversion around 3.80%, which acts like a weak reflecting barrier. The expected changes are only $+0.18$ and $+0.63$ basis points at 5% and 8%, so the short rate is essentially a random walk in this regime. Gray's economic point is that mean reversion operates only occasionally, in high-rate, high-volatility episodes.
- **Persistence.** The staying probabilities imply expected durations of 31 weeks (regime 1) and 105 weeks (regime 2), with stationary probability $\pi_1 = 0.23$ and Markov persistence $p_{11} + p_{22} - 1 = 0.96$ *(computed)*.
- **Fit.** Switching alone removes almost all serial correlation in squared standardised residuals. The single-regime model's volatility is too high in calm periods and too low in turbulent ones.

- **Testing for a second regime.** The likelihood-ratio statistic is 1,148.51 *(computed)*, but it is not χ^2 because the second regime's parameters are unidentified under the null. Gray mentions Hansen's (1992) standardised LR test as burdensome and treats the statistic as informal evidence.
- **Dating.** The high-volatility regime covers 1973–75, 1979–83 and a short spike after the 1987 crash.
- **Limitation.** With constant variances, the conditional standard deviation simply mimics the regime probabilities. This motivates step 2.

5.3 Step 2: GARCH within regimes (Table 3)

	Single-regime GARCH		Regime-switching GARCH	
Parameter	Est.	t	Est.	t
λ_{01}; λ_{11}	0.0059; −0.0013		0.1407; −0.0141	
λ_{02}; λ_{12}	—		−0.0011; 0.0006	
ω_1	0.0004	1.14	0.1870	2.23
α_1	0.2095	2.55	0.4609	1.99
β_1	0.8208	13.46	0.1977	1.24
ω_2	—		0.0099	3.86
α_2	—		0.1655	2.76
β_2	—		0.2685	2.82
p_{11}; p_{22}			0.9739; 0.9896	
log-likelihood	129.73		151.23	
Jarque–Bera; $Q^2(15)$ [p]	407.16; 21.57	[0.120]	—; 4.01	[0.998]

Single-regime GARCH.

- $\hat{\alpha} + \hat{\beta} = 1.030$ *(computed)*, which violates covariance stationarity.
- An LR test rejects IGARCH ($\alpha + \beta = 1$): statistic 7.07, χ^2_1.
- GARCH beats the single-regime constant variance by an LR of 1,185.75 *(computed)*.
- Diagnostics remain poor: there is residual ARCH at short lags and strong non-normality.

Gray's averaging argument. Suppose an ARCH(1) coefficient is 0.7 in one regime and 0.3 in the other, and a single-regime model estimates 0.5. Then the error in the squared-shock equation is about $+0.2\,\varepsilon^2_{t-1}$ in the first regime and $-0.2\,\varepsilon^2_{t-1}$ in the second. Either way the squared residuals stay serially correlated, so parameter instability across regimes shows up as misfit in a single-regime model.

Regime-switching GARCH.

- **Regime 1 (high volatility).** Long-run mean 9.98% *(computed)*; the expected change at 15% is −7.08 basis points.
- **Regime 2 (low volatility).** Long-run mean 1.83% with negative reversion; the expected change at 8% is only +0.37 basis points.
- **Shock sensitivity and memory.** The high-volatility regime reacts more to shocks ($\alpha_1 = 0.46 > \alpha_2 = 0.17$) but remembers them less ($\beta_1 = 0.20 < \beta_2 = 0.27$).
- **Persistence within regimes.** $\alpha_1 + \beta_1 = 0.66$ and $\alpha_2 + \beta_2 = 0.43$ *(computed)*, against 1.03 in the single-regime model. Both regimes are stationary.
- **Regime persistence.** Expected durations are 38 and 96 weeks, $\pi_1 = 0.28$, and Markov persistence is 0.96 *(computed)*.
- **Tests.**

  - GARCH within regimes versus constant variance within regimes: LR 80.24, χ^2_4, significant.
  - Regime-switching versus single-regime GARCH: quasi-LR 43.00. Five parameters are unidentified under the null, so Gray appeals to economic rather than statistical significance.

- **Volatility path.** Conditional variances no longer just mimic regime probabilities; the Fed-experiment volatility stands out as larger than other high-volatility episodes.

Impulse responses. Gray traces the effect of a 1-point shock at t on the conditional variance at $t+1, \ldots, t+5$:

	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$	total $\alpha/(1-\beta)$ *(computed)*
Single-regime GARCH	0.2095	0.1720	0.1411	0.1159	0.0951	1.17
High-volatility regime	0.4609	0.0911	0.0180	0.0036	0.0007	0.57
Low-volatility regime	0.1655	0.0444	0.0119	0.0032	0.0009	0.23

Where these numbers come from

Every entry equals $\alpha_k \beta_k^{m-1}$ for $m = 1, \ldots, 5$ *(computed)*. These are the ARCH(∞) weights of each regime's GARCH recursion, as in Bollerslev (1986). So the table shows the response *if the process stayed in that regime and if the lagged variance were that regime's own variance.* In Gray's model the lagged variance is actually the regime-aggregated σ^2_{t-1}, so the true response also depends on the regime probabilities. The table is an interpretive device, not the model's exact dynamics, and this is exactly the ambiguity Haas, Mittnik & Paolella (2004) later criticise.

Two sources of persistence (Gray's own statement). If one regime has a high average variance, the other a low one, and both are persistent, volatility will be persistent even when individual shocks die out quickly. A single-regime GARCH "cannot capture the persistence of regimes", so "all of the persistence in volatility is thrown into the persistence of an individual shock". This is the clearest early statement of the thesis's central question.

5.4 Step 3: level effects and time-varying probabilities (Table 4)

	Single regime		Constant p_{kk}		Time-varying $p_{kk,t}$	
	Est.	t	Est.	t	Est.	t
α_1	0.2269	2.58	0.3038	1.74	0.2390	1.21
β_1	0.7893	11.15	0.0936	0.69	0.0987	−0.35
ψ_1 (level)	0.0133	3.32	0.1550	6.16	0.1726	2.50
α_2			0.1528	2.70	0.1744	3.37
β_2			0.2383	2.36	0.3474	0.64
ψ_2 (level)			0.0420	8.15	0.0327	0.79
p_{11} or c_1			0.9688		−0.4384	−0.12
d_1					0.2011	0.61
p_{22} or c_2			0.9881		2.1989	1.38
d_2					−0.0421	−0.24
log-likelihood	136.03		176.47		182.67	

Mean parameters omitted (all insignificant, similar to Table 3). One t-statistic for β_1 is printed as negative in the paper although the estimate is positive; reproduced as printed.

The regimes become qualitatively different.

- Regime 1 is a "CIR regime": the GARCH terms are insignificant and the level term is highly significant.

- Regime 2 is a "GARCH regime": both GARCH terms are significant, and although the level term is statistically significant it is economically negligible ($\psi_2^2 R_{t-1} \approx 0.009$ at 5% against a typical variance of about 0.12).
- **Persistence.** Within-regime persistence is below 0.4 in both regimes with constant probabilities (0.40 and 0.39 *(computed)*), against 1.016 *(computed)* in the single-regime model.
- **Three sources of clustering.** GARCH effects, persistent regimes with different variance levels, and persistent interest-rate levels feeding the level effect.

Time-varying transition probabilities.

- **Joint significance.** The LR against constant probabilities is 12.39, χ^2_2, significant at 1%, even though each c_k, d_k is individually insignificant.
- **Direction.** $d_1 > 0$, so the high-volatility, mean-reverting regime is *more* persistent when rates are high. $d_2 < 0$, so the low-volatility random-walk regime is *less* persistent when rates are high. Rising rates make a switch into the mean-reverting regime likelier, which stops the random-walk regime from "wandering off".

Rate level R_{t-1} (%)	3	5	8	10	12	15
$p_{11,t} = \Phi(c_1 + d_1 R)$	0.565	0.715	0.879	0.942	0.976	0.995
expected duration, regime 1 (weeks)	2.3	3.5	8.3	17.3	41.4	201
$p_{22,t} = \Phi(c_2 + d_2 R)$	0.981	0.977	0.969	0.962	0.955	0.941
expected duration, regime 2 (weeks)	52	43	32	27	22	17

All entries *(computed)* from the Table 4 estimates; not reported in the paper.

The full GRS model against its single-regime version gives a quasi-LR of 93.28 (non-standard). Gray again argues for the second regime on economic grounds.

5.5 Out-of-sample tests (Table 6)

Parameters are estimated on one period and held fixed; conditional variances are then computed over a later period and compared with squared innovations ε_t^2. Three loss functions are used: RMSE, MAE and

$$R^2 = 1 - \frac{\sum_t (\varepsilon_t^2 - \hat{\sigma}_t^2)^2}{\sum_t \varepsilon_t^4}. \tag{11}$$

This R^2 imposes intercept 0 and slope 1 on the regression of realised on forecast volatility, so it judges the forecast itself, and it can be negative.

In sample					Out of sample				
Period		Const.	GARCH	GRS	Period		Const.	GARCH	GRS
1970–78	RMSE	.1847	.1857	**.1835**	1979–94[a]	RMSE	.7760	.7895	**.7625**
(470)	MAE	.0805	.0787	**.0711**	(796)	MAE	**.1704**	.1990	.1778
	R^2	.1136	.1253	**.1439**		R^2	.0261	.0352	**.0862**
1970–81	RMSE	.8362	.8682	**.8127**	1982–94	RMSE	.3332	.3294	**.3172**
(626)	MAE	.2623	.2554	**.2215**	(640)	MAE	.1885	**.0859**	.1244
	R^2	.0486	.0003	**.1115**		R^2	−.1511	−.0529	**.0471**
1976–82	RMSE	1.1233	1.2080	**.5528**	1983–94	RMSE	.3041	.1414	**.1363**
(365)	MAE	.4264	.4396	**.3737**	(488)	MAE	.2687	.0602	**.0583**
	R^2	.0709	−.0212	**.1279**		R^2	−4.3375	−.1311	**.0419**
1970–91	RMSE	.6686	.6870	**.6378**	1991–94	RMSE	.1287	.0184	**.0178**
(1110)	MAE	.1924	.1807	**.1561**	(156)	MAE	.1274	**.0137**	.0140
	R^2	.0405	.0098	**.1351**		R^2	−36.94	−.0053	**.0370**

Bold marks the best model in each row *(computed)*. Observation counts in parentheses. [a]Printed as "January 1979–December 1994" although the sample ends in April 1994. "Const." is the single-regime constant-variance model, "GARCH" the single-regime GARCH(1,1), "GRS" the full model with time-varying probabilities.

Reading the table.

- **Design of the four tests.** Test 1 forecasts the Fed experiment from data that include only the oil shock. Test 2 forecasts a calmer second half from a volatile first half. Test 3 shows that one episode of each regime suffices for identification. Test 4 forecasts the calm final three years, whose variance is 17 times lower than the estimation period's.
- **Negative R^2.** This arises when the estimation period is much more volatile than the forecast period, so the forecasts are systematically too high (the constant-variance model reaches -36.9).
- **Overall verdict.** Gray concludes that the GRS model "performs well in all four tests". By my count it has the lowest RMSE and the highest R^2 in all eight rows, but the lowest out-of-sample MAE in only one of four tests (single-regime GARCH wins two, constant variance one) *(computed)*.

5.6 Practical uses (section 5.3)

Gray suggests value-at-risk, Monte Carlo pricing of interest-rate securities (starting simulations from the current regime probabilities), and the risk that single-regime models misprice options by averaging regimes. He also notes that if the regime means are similar, the mixture is unimodal and can be approximated by a single normal with variance $\sum_k \xi_{k,t|t-1}\sigma^2_{k,t}$.

6 Assessment for the thesis

What Gray gains and what it costs

Gains.

- Full GARCH dynamics in each regime, with *all* parameters switching.
- A first-order recursive likelihood that is as easy to compute as GARCH.
- Clear evidence that within-regime persistence is far below single-regime persistence, while regimes themselves are persistent.

Costs (several argued in later papers in the folder):

- **Interpretation** (Haas et al., 2004). Because (5) feeds the aggregated σ^2_{t-1} into every regime, β_k is not the memory of regime k, and regime variances move with the filtered probabilities even if all $\alpha_k = 0$. The impulse-response table above should be read with this in mind.
- **No stationarity theory** for the collapsed model.
- **Not an estimator of the path-dependent model.** Augustyniak (2014) shows by simulation that Gray's MLE is biased for the parameters of (1), and the bias does not vanish with T.
- **Multi-step forecasting is awkward** (Klaassen, below).
- **Formal tests.** The second regime is never formally tested; the Davies problem is acknowledged.

Relevance to S&P 500 returns. The level effect and time-varying probabilities are specific to interest rates and need not carry over. The core collapsing idea (4)–(5) does carry over, and with zero-mean regimes it simplifies to $\sigma^2_{t-1} = \sum_j \xi_{j,t-1|t-2}\,\sigma^2_{j,t-1}$.

Part II

Klaassen (2002)

7 The question

GARCH(1,1) is the standard volatility forecasting tool, and Andersen & Bollerslev (1998) argue that its forecasts are good. Klaassen shows that, for daily dollar exchange rates, GARCH forecasts are nonetheless **too high in volatile periods**. He attributes this to the high persistence of individual shocks in GARCH. Following Lamoureux & Lastrapes (1990, *JBES*) and Timmermann (2000), he argues that a variance that alternates between high and low homoskedastic periods already produces persistence; a GARCH model, which cannot represent such periods, loads all of it onto individual shocks.

The paper's aim is practical: better volatility forecasts. Its theoretical contribution is a regime-switching GARCH specification for which *multi-step* variance forecasts are a simple recursion.

8 The model

8.1 Mean, regimes and distribution

Let $r_t = 100(\ln X_t - \ln X_{t-1})$ be the percentage depreciation of the dollar, where X_t is the spot rate. The mean is constant (a random walk with drift):

$$r_t = \mu + \varepsilon_t. \tag{12}$$

An AR(1) term is added for sterling. The regime $S_t \in \{1, 2\}$ follows a first-order Markov chain with constant staying probabilities p_{11}, p_{22}, where **regime 1 is the low-variance regime**. Conditional on observables and the regime path, ε_t is Student-t with ν degrees of freedom, mean zero and variance equal to the regime variance. Klaassen motivates the t explicitly: under normality, a single large outlier in a calm period would be read as a switch to the high-volatility regime, so the t "enhances the stability of the regimes".

8.2 Four ways to specify the regime variance

Klaassen lays out the options in a sequence; the comparison is the clearest statement of the collapsing branch in the folder.

1. **Path dependent** (his eq. 3): $\sigma_t^2(S_{1:t}) = \omega_{S_t} + \alpha_{S_t}\varepsilon_{t-1}^2 + \beta_{S_t}\sigma_{t-1}^2(S_{1:t-1})$. This is infeasible, because the number of paths grows exponentially.
2. **Cai; Hamilton & Susmel.** Drop the GARCH term, so the variance depends on only a few recent regimes.
3. **Gray** (his eq. 4). Integrate out S_{t-1} *inside* the lagged variance, using information up to $t-2$:

$$\sigma_{k,t}^2 = \omega_k + \alpha_k\varepsilon_{t-1}^2 + \beta_k \sum_j \xi_{j,t-1|t-2}\, \sigma_{j,t-1}^2. \tag{13}$$

   With a common mean, as here, this is exactly Gray's (5), because the mean-dispersion term in (4) vanishes.

4. **Klaassen** (his eq. 5). Postpone the integration to $t-1$ and condition on the current regime:

$$\boxed{\sigma_{k,t}^2 = \omega_k + \alpha_k\varepsilon_{t-1}^2 + \beta_k \sum_j w_{jk,t}\, \sigma_{j,t-1}^2, \qquad w_{jk,t} \equiv P(S_{t-1} = j \mid \mathcal{F}_{t-1}, S_t = k).} \tag{14}$$

Positivity requires $\omega_k > 0$ and $\alpha_k, \beta_k \geq 0$. If all regime parameters are equal, the model reduces to standard GARCH(1,1).

8.3 The weights: derivation

By Bayes' rule,

$$w_{jk,t} = \frac{P(S_t = k \mid S_{t-1} = j)\, P(S_{t-1} = j \mid \mathcal{F}_{t-1})}{P(S_t = k \mid \mathcal{F}_{t-1})} = \frac{p_{jk}\, \xi_{j,t-1|t-1}}{\xi_{k,t|t-1}}, \qquad \xi_{k,t|t-1} = \sum_i p_{ik}\, \xi_{i,t-1|t-1}. \tag{15}$$

Note that $P(S_{t-1} = j \mid \mathcal{F}_{t-1})$ is the *filtered* probability: $\mathcal{F}_{t-1}$ already contains r_{t-1}. This is the content of Klaassen's eq. (31), $p_{t-1}(r_{t-1}) = p_{t-2}(r_{t-1} \mid s_{t-1})$.

Gray versus Klaassen: the only difference is the weights

$$\text{Gray:} \quad \sigma^2_{k,t} = \omega_k + \alpha_k \varepsilon^2_{t-1} + \beta_k \sum_j \underbrace{\xi_{j,t-1|t-2}}_{\text{ex ante, same for all } k}\, \sigma^2_{j,t-1}$$

$$\text{Klaassen:} \quad \sigma^2_{k,t} = \omega_k + \alpha_k \varepsilon^2_{t-1} + \beta_k \sum_j \underbrace{\frac{p_{jk}\, \xi_{j,t-1|t-1}}{\xi_{k,t|t-1}}}_{\text{filtered, depends on } k}\, \sigma^2_{j,t-1}$$

- **More data.** Klaassen's weight uses r_{t-1}, which is informative about the regime at $t-1$.
- **The current regime.** It uses the hypothesis $S_t = k$, which with persistent regimes is very informative about S_{t-1}.

Gray's weights are the same for both regimes; Klaassen's differ by regime.

Why this pulls Klaassen's model towards IS-GARCH

From (15), the weight on the regime's own lagged variance is

$$w_{kk,t} = \frac{p_{kk}\, \xi_{k,t-1|t-1}}{p_{kk}\, \xi_{k,t-1|t-1} + p_{jk}\, \xi_{j,t-1|t-1}}, \qquad j \neq k.$$

With persistent regimes (p_{jk} small) this is close to one unless the filter is very sure the process was in the other regime. For example, with $p_{kk} = 0.99$, $p_{jk} = 0.01$ and an uninformative filter ($\xi = 0.5$ each), $w_{kk,t} = 0.99$. So in practice each regime's variance mostly feeds on its *own* lag, which is precisely the structure Haas, Mittnik & Paolella (2004) impose exactly. Gray's weights, by contrast, mix the two regime variances in the same proportions for both regimes. This is useful for Chapter 3: the collapse and remove branches are closer in Klaassen's version than in Gray's.

8.4 Why the tree recombines

Because the sum in (14) runs over S_{t-1} with weights that depend only on observables and on S_t, $\sigma^2_{k,t}$ depends on the current regime alone. Klaassen notes that integrating over the whole path $S_{1:t-1}$ is equivalent to integrating over S_{t-1} only, because the lagged $\sigma^2_{j,t-1}$ is already path-free by the same construction one period earlier.

9 Properties

9.1 Flexibility in volatility persistence

- **Two routes to a volatile period.** A shock can be followed by a volatile period through GARCH effects or through a switch to the high-variance regime.
- **Regime-specific persistence.** Shocks can be more persistent in one regime than the other.
- **"Pressure-relieving" shocks.** A large shock followed by calm is captured in two ways: a switch to the low-variance regime and, if that regime also has low persistence, fast forgetting of the shock.
- **Contrast with Branch 1.** In Cai and Hamilton & Susmel the regimes differ only by an additive or multiplicative constant, not by their ARCH parameters.

9.2 Recursive multi-step forecasting (the main contribution)

Let $r_{t,h} = r_t + \cdots + r_{t+h-1}$ be the h-day change. Because one-day changes are serially uncorrelated (eq. 7),

$$\mathrm{Var}(r_{t,h} \mid \mathcal{F}_{t-1}) = \sum_{\tau=t}^{t+h-1} \mathrm{Var}(r_\tau \mid \mathcal{F}_{t-1}), \qquad \mathrm{Var}(r_\tau \mid \mathcal{F}_{t-1}) = \sum_k \xi_{k,\tau|t-1}\, \sigma^2_{k,\tau|t-1} \quad \text{(eq. 8)}, \tag{16}$$

where $\sigma^2_{k,\tau|t-1} \equiv \mathrm{Var}(\varepsilon_\tau \mid \mathcal{F}_{t-1}, S_\tau = k)$. Two ingredients are needed.

(a) Future regime probabilities come from powers of the transition matrix (App. A, eqs. 15–17):

$$\xi_{\cdot,\tau|t-1} = (P')^{\tau-t}\, \xi_{\cdot,t|t-1}.$$

Klaassen's "M" is our P': his matrix is column-stochastic.

(b) Future regime variances follow the recursion (eq. 9):

$$\boxed{\sigma^2_{k,\tau|t-1} = \omega_k + (\alpha_k + \beta_k) \sum_j P(S_{\tau-1} = j \mid \mathcal{F}_{t-1}, S_\tau = k)\, \sigma^2_{j,\tau-1|t-1}, \qquad \tau > t,} \tag{17}$$

started from $\sigma^2_{k,t|t-1} = \sigma^2_{k,t}$ (known at $t-1$). The backward probabilities again come from Bayes' rule (eq. 22),

$$P(S_{\tau-1} = j \mid \mathcal{F}_{t-1}, S_\tau = k) = \frac{p_{jk}\, \xi_{j,\tau-1|t-1}}{\xi_{k,\tau|t-1}}.$$

Proof idea (App. A). Take conditional expectations of (14) dated τ, given $(\mathcal{F}_{t-1}, S_\tau = k)$, and apply the law of iterated expectations to each term.

- **ARCH term.** $E[\varepsilon^2_{\tau-1} \mid \mathcal{F}_{t-1}, S_\tau] = E\big[\mathrm{Var}(\varepsilon_{\tau-1} \mid \mathcal{F}_{t-1}, S_{\tau-1}) \mid \mathcal{F}_{t-1}, S_\tau\big]$. This uses the fact that, given the regime at $\tau-1$, the error at $\tau-1$ does not depend on the regime at τ.
- **GARCH term.** The same steps show that the expected lagged weighted variance equals the same object. The key step is that, given $S_{\tau-1}$, earlier data are independent of S_τ (Markov property).

Both terms therefore collapse to $\sum_j P(S_{\tau-1} = j \mid \cdot)\, \sigma^2_{j,\tau-1|t-1}$, and their coefficients add to $\alpha_k + \beta_k$.

Check. If all regime parameters are equal, (17) becomes $\sigma^2_{\tau|t-1} = \omega + (\alpha+\beta)\sigma^2_{\tau-1|t-1}$, the standard GARCH forecast recursion.

Why the recursion works for Klaassen but not for Gray

The paper says only that multi-step forecasts with Gray's model "turn out to be very complicated". My reading of why:

- **Klaassen.** Equation (14) defines $\sigma^2_{k,t}$ as an expectation conditional on $(\mathcal{F}_{t-1}, S_t = k)$. That conditioning set is exactly the one the tower property needs to chain one horizon to the next; the only probabilities involved are transition probabilities and probabilities propagated with P', all known at $t-1$.
- **Gray.** In Gray's model, the variance two steps ahead uses the aggregate $\sigma^2_t = \sum_j \xi_{j,t|t-1}\sigma^2_{j,t}$ of the next period, and later steps use the weights $\xi_{\cdot,t+1|t}$, which depend on the not-yet-observed r_t through Bayes' rule. Their expectation is a nonlinear function of future data, so E_{t-1} does not pass through, and there is no closed form. This matches what Klaassen reports in practice: he could not compute Gray's ten-day forecasts at all.

9.3 The "unconditional" regime variances

Assume that $\bar\sigma^2_k \equiv \mathrm{Var}(\varepsilon_t \mid S_t = k)$ exists and does not depend on t. Taking unconditional expectations of (14) given $S_t = k$, with the same law-of-iterated-expectations steps (App. B, eq. 23), gives

$$\bar\sigma^2_k = \omega_k + (\alpha_k + \beta_k) \sum_j P(S_{t-1} = j \mid S_t = k)\, \bar\sigma^2_j.$$

Stacking the two regimes (eqs. 10 and 24):

$$\begin{pmatrix} \bar\sigma^2_1 \\ \bar\sigma^2_2 \end{pmatrix} = (I_2 - \mathbf{A})^{-1} \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}, \qquad \mathbf{A}_{kj} = P(S_{t-1} = j \mid S_t = k)\,(\alpha_k + \beta_k), \tag{18}$$

where the backward probabilities come from Bayes' rule with the stationary probabilities π_j (eqs. 26–27): $P(S_{t-1} = j \mid S_t = k) = p_{jk}\pi_j/\pi_k$. Klaassen derives three **necessary** conditions for both variances to exist:

$$\mathbf{A}_{11} < 1, \qquad \mathbf{A}_{22} < 1, \qquad \det(I_2 - \mathbf{A}) > 0.$$

They parallel $\alpha + \beta < 1$ in GARCH(1,1), where $\bar\sigma^2 = \omega/(1-\alpha-\beta)$. Table 1 reports $\bar\sigma^2_k$ instead of ω_k to make the models easier to compare.

Two simplifications (my derivations)

1. For two regimes the backward probabilities equal the forward ones. With $\pi_1 = (1-p_{22})/(2-p_{11}-p_{22})$, one checks $\pi_1 p_{12} = \pi_2 p_{21}$; every stationary two-state chain is reversible. Hence $P(S_{t-1} = j \mid S_t = k) = p_{jk}\pi_j/\pi_k = p_{kj}$, so

$$\mathbf{A} = \mathrm{diag}(\alpha_1 + \beta_1,\ \alpha_2 + \beta_2)\, P.$$

2. The three conditions amount to $\rho(\mathbf{A}) < 1$. For a nonnegative 2×2 matrix, $1 - \mathbf{A}_{11} > 0$ together with $\det(I - \mathbf{A}) > 0$ is the standard condition for $I - \mathbf{A}$ to be a nonsingular M-matrix, i.e. $\rho(\mathbf{A}) < 1$. This has the same form as the covariance-stationarity condition $\rho(\mathrm{diag}(a)P) < 1$ for Markov-switching ARCH (Francq et al., 2001, as cited in Haas et al., 2004). It is a convenient way to state Klaassen's condition in Chapter 3, provided it is flagged as a derivation. Klaassen proves *necessity* only: no stationarity theorem for the model exists in the folder.

Worked example (Deutschmark, regime-switching GARCH). Table 1 gives $p_{11} = 0.981$, $p_{22} = 0.998$; regime 1 has constant variance $\bar\sigma^2_1 = 0.12$ (no ARCH/GARCH terms, so $\alpha_1 + \beta_1 = 0$); regime 2 has $\alpha_2 = 0.07$, $\beta_2 = 0.90$ and $\bar\sigma^2_2 = 0.57$. Then $\mathbf{A}_{21} = 0.002 \times 0.97 = 0.0019$ and $\mathbf{A}_{22} = 0.998 \times 0.97 = 0.968$, and from (18) the implied intercepts are $\omega_1 = 0.12$ and $\omega_2 = 0.57(1 - 0.968) - 0.0019 \times 0.12 \approx 0.018$ *(computed)*. The condition $\mathbf{A}_{22} < 1$ holds comfortably even though the high-variance regime's persistence is 0.97.

9.4 The likelihood

The density of r_t given observables is a two-component t mixture (App. C, eq. 28):

$$f(r_t \mid \mathcal{F}_{t-1}) = \sum_k \xi_{k,t|t-1}\, f_{k,t}, \qquad f_{k,t} = t_\nu\big(r_t;\, \mu,\, \sigma^2_{k,t}\big), \qquad \ell_T(\theta) = \sum_t \ln f(r_t \mid \mathcal{F}_{t-1}),$$

with the filtered probability updated by Bayes' rule (eq. 31): $\xi_{k,t|t} = \xi_{k,t|t-1} f_{k,t} / f(r_t \mid \mathcal{F}_{t-1})$. Like Gray's, the likelihood is first-order recursive.

Klaassen's filter, one period at a time (thesis notation)

Carry forward $\xi_{j,t-1|t-1}$, $\sigma^2_{j,t-1}$ $(j = 1, 2)$ and $\varepsilon_{t-1} = r_{t-1} - \mu$.

1. **Predict:** $\xi_{k,t|t-1} = \sum_j p_{jk}\, \xi_{j,t-1|t-1}$.
2. **Backward weights:** $w_{jk,t} = p_{jk}\, \xi_{j,t-1|t-1} / \xi_{k,t|t-1}$ (four numbers).
3. **Regime variances:** $\sigma^2_{k,t} = \omega_k + \alpha_k \varepsilon^2_{t-1} + \beta_k \sum_j w_{jk,t}\, \sigma^2_{j,t-1}$.
4. **Densities:** $f_{k,t}$ (Student-t, mean μ, variance $\sigma^2_{k,t}$); add $\ln \sum_k \xi_{k,t|t-1} f_{k,t}$ to ℓ_T.
5. **Update:** $\xi_{k,t|t} = \xi_{k,t|t-1} f_{k,t} \big/ \sum_i \xi_{i,t|t-1} f_{i,t}$.

Start-up. Klaassen sets the required variables to their unconditional expectations: the variances to $\bar{\sigma}^2_k$ from (18), and presumably the probabilities to π_k [VERIFY: the paper's wording is general].

Gray's version. Gray differs only at steps 2–3: every regime uses $\sum_j \xi_{j,t-1|t-2} \sigma^2_{j,t-1}$.

9.5 Regime inference and durations

Ex ante probabilities come out of the filter. Smoothed probabilities $\xi_{k,t|T}$ are computed forward from them with Gray's 1996b algorithm (App. D). The expected duration of regime 2 is (eq. 11)

$$\sum_{h \geq 1} h\, p_{22}^{h-1}(1 - p_{22}) = \frac{1}{1 - p_{22}},$$

so a staying probability of 0.98 means 50 trading days, about 2.5 months.

10 In-sample results

10.1 Data and model labels

Daily dollar prices of the British pound, German mark and Japanese yen, 3 January 1978–23 July 1997 (4,982 changes; Datastream). The paper writes GARCH$(P_1, Q_1; P_2, Q_2)$ for a regime-switching model with Q_k ARCH and P_k GARCH terms in regime k. So GARCH(0,2;1,1) has an ARCH(2) low-variance regime and a GARCH(1,1) high-variance regime, and ARCH$(Q_1; Q_2)$ has only ARCH terms. Six models are estimated for each currency:

1. constant variance;
2. GARCH(1,1);
3. ARCH(0;0), i.e. switching with a constant variance in each regime;
4. ARCH(Q_1;4), in the spirit of Cai and Hamilton–Susmel but less restricted;
5. Gray's variant;
6. Klaassen's regime-switching GARCH.

10.2 Table 1: estimates

			Low-variance regime 1			High-variance regime 2					
	Model	ν^{-1}	$\bar{\sigma}_1^2$	ARCH	p_{11}	$\bar{\sigma}_2^2$	α_2	ARCH 2–4	β_2	p_{22}	$\Delta\ell$
GBP	GARCH(1,1)	.19				.61	.06		.93		−4437.26
	ARCH(0;0)	.17	.18		.984	.65				.987	−46.53
	ARCH(2;4)	.19	.16	.11, .11	.991	.58	.07	.04, .07, .07		.996	−19.56
	Gray(0,2;1,1)	.19	.18	.21, .18	.999	.74	.06		.93	.999	+9.24
	GARCH(0,2;1,1)	.18	.15	.17, .10	.989	.58	.05		.93	.997	+19.34
DEM	GARCH(1,1)	.20				.70	.08		.91		−4778.34
	ARCH(0;0)	.15	.23		.984	.77				.982	−24.85
	ARCH(0;4)	.16	.21		.983	.70	.06	.01, .07, .05		.987	−14.97
	Gray(0,0;1,1)	.20	.11		.998	.70	.08		.89	.999	−3.29
	GARCH(0,0;1,1)	.19	.12		.981	.57	.07		.90	.998	+8.44
JPY	GARCH(1,1)	.25				.61	.09		.87		−4682.58
	ARCH(0;0)	.20	.21		.975	.73				.975	+9.07
	ARCH(1;4)	.20	.23	.08	.981	.71	.03	.04, .05, .00		.981	+16.28
	Gray(0,1;1,1)	.21	.23	.10	.980	.80	.04		.48	.980	+12.59
	GARCH(0,1;1,1)	.21	.22	.09	.978	.68	.06		.78	.984	+16.52

Standard errors and the constant-variance rows omitted; means all ≈ 0. For GARCH(1,1) the $\bar{\sigma}^2$ column is the single-regime unconditional variance and α_2, β_2 are α, β. $\Delta\ell$ is the log-likelihood minus that of GARCH(1,1); for GARCH(1,1) it is the log-likelihood itself. "ARCH" lists the lag-1 (and lag-2) ARCH coefficients of regime 1.

Step by step.

- **Single-regime GARCH(1,1).** It beats constant variance by a huge margin (e.g. 244.34 log points for sterling), and GARCH(2,1) and GARCH(1,2) add nothing. Persistence is $\alpha + \beta =$ 0.99, 0.99, 0.96 *(computed)*.
- **ARCH(0;0): regimes only.** The high-variance regime's variance is 3.3–3.6 times the low one's *(computed)*. All staying probabilities exceed 0.975 (expected durations 40–77 days *(computed)*). For the yen this model already beats GARCH (+9.07), but squared residuals still show clustering (Table 2), so there is heteroskedasticity *within* regimes.
- **ARCH(Q_1;4): ARCH within regimes.** Likelihood-ratio tests cut the low-variance regime to 2 lags (sterling; 0.74 [0.69]), 0 lags (mark; 1.87 [0.76]) and 1 lag (yen; 3.79 [0.29]). Cutting the high-variance regime to match is rejected for sterling and the mark (23.71 and 19.76, both [0.00]) and marginally for the yen (7.06 [0.07]). So **there is more volatility clustering in the high-variance regime**, which contradicts the symmetric structure of Cai and Hamilton & Susmel.
- **Adding a GARCH term to the high regime.** The log-likelihood rises by 38.90 (sterling), 23.41 (mark) and 0.24 (yen) with *two fewer* parameters, and residual heteroskedasticity disappears. Against Cai-style and Hamilton–Susmel-style fourth-order ARCH models the gains are 40.46 / 24.93 / 0.34 and 43.12 / 23.14 / 1.10 respectively.
- **Klaassen versus Gray.** Similar estimates, but Klaassen's likelihood is higher by 10.10, 11.73 and 3.93, and Gray's model leaves significant residual heteroskedasticity for the mark ($Q_{10} =$ 27.08, $p = 0.00$). Klaassen attributes this to Gray's less efficient use of conditioning information.
- **Klaassen versus GARCH(1,1).** Gains of 19.34, 8.44 and 16.52 log points. Klaassen notes that the specifications used do not strictly nest GARCH(1,1), because the low regime has no GARCH term.
- **Role of the t distribution.** With normal errors, sterling's smoothed probabilities show many more regime switches (his Fig. 1C).

- **Smoothed probabilities.** The European currencies show few regime shifts (smooth GARCH transitions dominate); the yen shows more, so sudden variance shifts matter more there.

What Table 1 implies about where persistence lives (computed)

	GARCH $\alpha+\beta$	high regime $\alpha_2+\beta_2$	dur. low	dur. high	π_2	$p_{11}+p_{22}-1$
GBP	0.99	0.98	91	333	0.79	0.986
DEM	0.99	0.97	53	500	0.90	0.979
JPY	0.96	0.84	45	63	0.58	0.962

Durations are in trading days and π_2 is the stationary probability of the high-variance regime.

- **Little change within the high regime for sterling and the mark.** Persistence there is almost as high as single-regime GARCH (0.98 and 0.97). Only for the yen does it fall noticeably (0.84).
- **The high regime is the normal state.** It is occupied 79–90% of the time for the European currencies. So the "high-variance" regime is better read as the normal GARCH state, and the low-variance regime as an occasional tranquil interlude (in which shocks are quickly forgotten).
- **Where the gain comes from.** This matches Klaassen's own decomposition below: the gain comes from *slightly* lower shock persistence in volatile periods plus a separate tranquil regime. It is not a wholesale transfer of persistence to the Markov chain.

For the thesis this is a useful counterpoint to Gray, where within-regime persistence collapses.

10.3 Where the improvement over GARCH comes from

- **When it outperforms.** Klaassen regresses $\mathbf{1}[\Delta\ell_t > 0]$ (the regime-switching model's log-likelihood contribution beats GARCH) on log average past volatility and its square, with t-values between 6 and 9. The relationship is U-shaped: the regime-switching model wins mainly in tranquil *and* in volatile periods, because its variance estimates are lower in both.
- **Volatile periods.** Re-estimating with the high regime's parameters fixed at the GARCH(1,1) values, $(\alpha_2, \beta_2) = (\hat{\alpha}, \hat{\beta})$, removes the outperformance there. The restriction is rejected for the mark and yen (not sterling), so the small differences matter. Conclusion: shocks have a smaller direct effect ($\hat{\alpha}_2 < \hat{\alpha}$) and a smaller cumulative effect ($\hat{\alpha}_2 + \hat{\beta}_2 < \hat{\alpha} + \hat{\beta}$).
- **Tranquil periods.** Adding the probability of the low-variance regime to the regression gives t-values of 7–11 and removes the effect of past volatility. So the tranquil-period gain comes from the low-variance regime.
- **Example.** The dollar's sharp fall after the G-5 Plaza announcement (22 September 1985) was followed by calm. GARCH keeps the variance high; the regime-switching model temporarily switches to the low regime and lets the variance fall quickly.

11 Out-of-sample forecasting

11.1 Design

- **Sample split.** The sample is split into two halves of 2,491 days; the second half starts on 20 October 1987. Models are estimated on one half with parameters fixed, forecasts made for the other half, and then the roles are reversed. This gives two estimates of everything.
- **Horizons.** One day and ten days ($h = 1, 10$).

- **Realised volatility.** Following Andersen & Bollerslev (1998), the sum of squared daily changes over the horizon (eq. 12):

$$v_{t,h} = \sum_{\tau=t}^{t+h-1} (r_\tau - \hat{\mu})^2.$$

  It is unbiased and, for $h > 1$, less noisy than the squared h-day change.

11.2 Forecast efficiency regressions

The forecast efficiency (Mincer–Zarnowitz-type) regression is (eq. 13)

$$v_{t,h} = \kappa_0 + \kappa_1 \widehat{\mathrm{Var}}(r_{t,h} \mid \mathcal{F}_{t-1}) + u_t, \qquad H_0 : \ \kappa_0 = 0, \ \kappa_1 = 1.$$

Standard errors are Newey–West (Bartlett kernel, automatic lag selection) and are multiplied by $\sqrt{2}$ to account for parameter estimation, following West & McCracken (1998) for the "fixed" scheme with equal in- and out-of-sample sizes.

- **GARCH(1,1).** In all 12 cases (3 currencies $\times$ 2 horizons $\times$ 2 splits) $\hat{\kappa}_0 > 0$ and $\hat{\kappa}_1 < 1$; in 8 of 12 both are significantly rejected.
- **Which forecasts are wrong.** Allowing a break at the median forecast gives average coefficients $(\hat{\kappa}_0, \hat{\kappa}_1)$ of $(0.00, 1.02)$ $(h = 1)$ and $(0.38, 1.09)$ $(h = 10)$ for below-median forecasts, never significantly different from $(0, 1)$. For above-median forecasts the averages are $(0.23, 0.55)$ and $(2.96, 0.43)$, significant in 9 of 12. **High GARCH forecasts overstate the variance; low ones are fine.**
- **Regime-switching models.** $\kappa_0 = 0$, $\kappa_1 = 1$ is "generally not rejected". A few significant slopes remain, e.g. the yen (0.65 at $h = 1$, 0.40 at $h = 10$ in one split).

11.3 MSE and restricted R^2 (Table 3)

The restricted R^2 is (eq. 14)

$$\tilde{R}^2 = 1 - \frac{\mathrm{Var}\big(v_{t,h} - \widehat{\mathrm{Var}}(r_{t,h} \mid \mathcal{F}_{t-1})\big)}{\mathrm{Var}(v_{t,h})}.$$

Two details of $\tilde{R}^2$ worth knowing

- **Constant forecasts score exactly zero.** Because the formula uses *variances*, any constant forecast gives $\tilde{R}^2 = 0$ exactly, which is why the constant-variance rows of Table 3 are all 0.
- **Constant bias is not penalised.** Subtracting a constant does not change a variance, so a forecast that is off by the same amount every day scores the same as an unbiased one. Klaassen calls $\tilde{R}^2$ "similar to" Gray's measure (11), but Gray's uses raw sums of squares and *does* penalise level bias; that is why Gray's can reach -36.9. Keep the two apart if both are used in Chapter 5.

		$h = 1$				$h = 10$			
		MSE−G(1,1)		$\tilde{R}^2$		MSE−G(1,1)		$\tilde{R}^2$	
GBP	GARCH(1,1)	.697	.998	.062	.058	11.20	16.39	.256	.226
	Gray	.002	−.002	.059	.059	—	—	—	—
	Klaassen	.001	−.004	.060	.061	−0.03	−0.56	.259	.253
DEM	GARCH(1,1)	.817	1.208	.040	.029	14.51	22.18	.122	.031
	Gray	−.004*	−.008	.046	.034	—	—	—	—
	Klaassen	−.009*	−.011*	.050	.036	−1.48*	−1.69*	.199	.095
JPY	GARCH(1,1)	.996	1.414	.009	.005	17.35	23.59	.052	−.096
	Gray	−.021*	−.022	.025	.019	—	—	—	—
	Klaassen	−.024*	−.010	.028	.011	−2.32*	−1.44	.117	.012

For GARCH(1,1) the MSE column is the MSE itself; other rows are differences from it (negative = better). Two columns per statistic: left = forecasts of the second half, right = of the first half. * significant at 5%. Gray's model cannot be used for $h = 10$. Table 3 labels the yen models "(1,0;1,1)" whereas Table 1 has "(0,1;1,1)"; this appears to be a typo.

Headline results.

- **MSE.** Klaassen's model has lower MSE than GARCH in **11 of 12** cases (6 significant). The exception is sterling, $h = 1$, second half (+.001).
- $\tilde{R}^2$**.** Higher than GARCH in 11 of 12 cases. The average improvement is 0.007 ($h = 1$) and 0.057 ($h = 10$), relative to GARCH averages of 0.034 and 0.099, so the **relative improvement is 22% and 58%**. Recomputing from the rounded table entries gives 21% and 58% *(computed).*
- **Versus Gray.** Klaassen beats Gray at $h = 1$ in 5 of 6 cases (the exception is the yen, first half).
- **Regime-switching ARCH versus GARCH.** Better in only 15 of 24 cases (4 significant), all for the mark and yen; for sterling GARCH wins all 8. So regimes *and* GARCH terms are both needed.
- **Why the $\tilde{R}^2$ values are so low.** This reflects noise in daily realised volatility, not bad forecasts. Andersen & Bollerslev show that the R^2 of GARCH rises from 0.036 to 0.436 when realised volatility is built from five-minute returns. It also explains why $\tilde{R}^2$ is higher at $h = 10$.

12 Klaassen's conclusions and caveats

- **Main findings.**
  - Regimes matter: even constant-variance regimes sometimes beat GARCH.
  - There is heteroskedasticity within regimes, and more of it in the high-volatility regime.
  - GARCH terms matter: single-regime GARCH sometimes beats multi-lag regime-switching ARCH.
  - The t distribution stabilises the regimes.
- **The regimes are a technical device.** Klaassen deliberately does *not* interpret the regime dates economically: regimes are "a technical means to obtain more flexibility regarding volatility persistence".
- **No formal test of the number of regimes.** Under the null the staying probabilities are unidentified (he cites Hansen, 1992), so the paper does not test for a second regime. It focuses on forecast quality instead.

What Klaassen gains and what it costs

Gains over Gray.

- Uses more information when collapsing (higher likelihood).
- Regime-specific weights that keep each regime's variance close to its own history.
- A GARCH-style recursion for multi-step forecasts.
- A closed form for the regime variances together with necessary existence conditions.

Costs shared with Gray.

- Still an approximation: not the path-dependent model.
- β_k still mixes in the other regime's variance with weight $w_{jk,t}$.
- No stationarity theorem.
- No asymptotic theory for the MLE in the folder.
- No formal regime test.

For Chapter 5.

- The forecast-efficiency regression, the $\sqrt{2}$ correction, realised-volatility sums and $\tilde{R}^2$ are a ready-made forecast evaluation design.
- The asymmetric specification (ARCH or constant low regime, GARCH high regime) is a reasonable restriction if the low-volatility GARCH parameters prove weakly identified on S&P 500 data, as they were in Bauwens et al. (2010).

Part III

Putting the two papers together

13 Side-by-side comparison

	Gray (1996)	Klaassen (2002)
Data	Weekly one-month T-bill rate changes, 1970–94 ($T = 1{,}267$)	Daily USD/GBP, DEM, JPY changes, 1978–97 ($T = 4{,}982$)
Regime 1	High volatility	Low variance
Means	Regime-specific, mean-reverting	Common constant (AR(1) for GBP)
Innovations	Normal	Student-t
Transition probabilities	Constant, or probit in the rate level	Constant
Variance	GARCH(1,1) + level effect in each regime	ARCH or constant low regime; GARCH(1,1) high regime
Lagged variance used	$\sum_j \xi_{j,t-1\|t-2}\, \sigma^2_{j,t-1}$ (+ mean dispersion)	$\sum_j w_{jk,t}\, \sigma^2_{j,t-1}$ with $w_{jk,t} = p_{jk}\xi_{j,t-1\|t-1}/\xi_{k,t\|t-1}$
Lagged shock	Centred on the mixture mean	Centred on the common mean
Information in weights	$\mathcal{F}_{t-2}$; same weights for both regimes	$\mathcal{F}_{t-1}$ and $S_t = k$; regime-specific weights
Likelihood	First-order recursive	First-order recursive
Multi-step forecasts	No closed form; not computed	GARCH-type recursion (17)
Stationarity theory	None	Closed-form $\bar{\sigma}^2_k$; necessary conditions only
Within-regime persistence	0.66 / 0.43 (vs 1.03 single regime)	0.98 / 0.97 / 0.84 in the high regime (vs 0.99 / 0.99 / 0.96)
Regime persistence	Staying probabilities 0.97 / 0.99	0.98–0.998
Regime test	Not formal (nuisance parameters)	Not attempted
Out-of-sample	Lowest RMSE and highest R^2 in all 4 tests; MAE mixed	Lower MSE than GARCH in 11/12; $\tilde{R}^2$ +22% / +58%

The thesis question, as these two papers answer it

Both papers support the claim that single-regime GARCH overstates the persistence of shocks, but by different amounts.

- **Gray (T-bill rates).** Within-regime persistence collapses from above one to well below one, so most persistence is attributed to the regimes.
- **Klaassen (exchange rates).** The dominant high-variance regime keeps persistence close to GARCH for sterling and the mark. The gain comes from a modest reduction in shock persistence plus a tranquil regime that lets the variance drop quickly.

The two are consistent with the synthesis in the main reading guide: where persistence ends up depends on the specification and on the data. In Klaassen's case it also depends on which regime is allowed GARCH dynamics, since he gives them only to the high regime.

14 Notes for hand-coding Klaassen's model in Chapter 5

This section is my implementation guidance, not content from the paper; the [VERIFY] items should be settled when the code is written.

1. **Filter.** Use the five-step filter in the keybox of Part II. For a zero-mean S&P 500 specification set $\mu = 0$ (MSGARCH also assumes demeaned data, so this keeps the models comparable). Gray's version is the same code with the weights $w_{jk,t}$ replaced by $\xi_{j,t-1|t-2}$ for both k, which makes it

cheap to estimate both as a check.

2. **Start-up.** Initialise $\sigma^2_{k,0}$ at $\bar{\sigma}^2_k$ from (18) (using the reversibility simplification $\mathbf{A} = \mathrm{diag}(\alpha_k + \beta_k)P$) and $\xi_{k,0|0}$ at π_k [VERIFY against the paper's wording].
3. **Constraints.** Impose $\omega_k > 0$, $\alpha_k, \beta_k \geq 0$, $\nu > 2$, $p_{kk} \in (0,1)$, and $\rho(\mathbf{A}) < 1$ so that (18) is defined for the start-up. Klaassen does not state how he imposed constraints [VERIFY].
4. **Label switching.** Fix regime 1 as the low-variance regime, e.g. by requiring $\bar{\sigma}^2_1 < \bar{\sigma}^2_2$.
5. **Asymmetric option.** Klaassen's best models have no GARCH term in the low regime. On S&P 500 data a GARCH(0,1;1,1) or GARCH(0,0;1,1) analogue is a natural fallback if the low regime's β_1 is weakly identified.
6. **Forecasts.** Implement (17) for h-day forecasts. Propagate $\xi_{\cdot,\tau|t-1}$ with P' and recompute the backward probabilities $p_{jk}\xi_{j,\tau-1|t-1}/\xi_{k,\tau|t-1}$ at each step. Validate against simulation: the mean of simulated h-day squared sums should match the recursion.
7. **Sanity check.** With $\omega_1 = \omega_2$, $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$, the filter must reproduce a single-regime GARCH(1,1) likelihood exactly.

15 Translation tables

Gray (1996)

Gray	This summary	Note
r_t; Δr_t	R_t; y_t	rate level and its weekly change (our r_t is a return)
Φ_{t-1}	$\mathcal{F}_{t-1}$	information set
$S_t \in \{1,2\}$	S_t	regime 1 = high volatility
$\mu_{it} = \alpha_i + \beta_i r_{t-1}$ (Tables: $a_{0i} + a_{1i} r_{t-1}$)	$\mu_{k,t} = \lambda_{0k} + \lambda_{1k} R_{t-1}$	avoids clash with α, β and a_m
h_{it}	$\sigma^2_{k,t}$	regime variance
h_t	σ^2_t	aggregate variance $\mathrm{Var}(y_t \mid \mathcal{F}_{t-1})$
ω_i, a_i, b_i (Tables: b_{0i}, b_{1i}, b_{2i})	$\omega_k, \alpha_k, \beta_k$	in Table 2, b_{0i} are standard deviations
σ^2_i (level term)	ψ^2_k	Tables report ψ_k
z_t	η_t	standardised innovation
P, Q; P_t, Q_t	p_{11}, p_{22}; $p_{11,t}, p_{22,t}$	staying probabilities
c_i, d_i	c_k, d_k	probit coefficients
$p_{1t} = \Pr(S_t = 1 \mid \Phi_{t-1})$	$\xi_{1,t|t-1}$	ex ante probability
$g_{it} = f(\Delta r_t \mid S_t = i)$	$f_{k,t}$	regime density

Klaassen (2002)

Klaassen	This summary	Note
S_t; s_t	X_t; r_t	log spot rate; 100× its change
m; e_t (also ε_t)	μ; ε_t	mean; innovation
$r_t \in \{1,2\}$; $\tilde{r}_t$	S_t; $S_{1:t}$	regime 1 = low variance; regime path
I_{t-1}	$\mathcal{F}_{t-1}$	observed information
$V_{t-1}\{e_t \mid \tilde{r}_t\}$	$\sigma_t^2(S_{1:t})$	path-dependent variance
$V_{t-1}\{e_t \mid r_t\}$	$\sigma_{k,t}^2$	regime variance
$p_{t-1}(r_t)$	$\xi_{k,t\|t-1}$	ex ante probability
$p_{t-1}(r_{t-1})$	$\xi_{j,t-1\|t-1}$	filtered (as I_{t-1} contains s_{t-1})
$p_{t-1}(r_{t-1} \mid r_t)$	$w_{jk,t}$	backward weight
$p_T(r_t)$	$\xi_{k,t\|T}$	smoothed probability
$\omega_r, \alpha_{qr}, \beta_r$	ω_k, $a_{m,k}$ or α_k, β_k	lag-m ARCH coefficient of regime k
M	P'	column-stochastic transition matrix (not the Haas M)
σ_r^2; A	$\bar{\sigma}_k^2$; $\mathbf{A}$	unconditional regime variance; matrix in (18)
n	ν	degrees of freedom; tables report ν^{-1}
$s_{t,h}$; $v_{t,h}$	$r_{t,h}$; $v_{t,h}$	h-day change; realised volatility
γ_0, γ_1	κ_0, κ_1	efficiency regression (γ is the Lyapunov exponent in the thesis)

16 Symbols used here that are not in notation.md

Local to this summary (needed only to describe Gray's interest-rate model; probably not needed in the thesis):

- **NEW SYMBOL** R_t: short-rate level.
- **NEW SYMBOL** y_t: generic observed series, here ΔR_t (also proposed for Hamilton, 1989).
- **NEW SYMBOL** $\lambda_{0k}, \lambda_{1k}$: mean-reversion coefficients.
- **NEW SYMBOL** ψ_k: level-effect coefficient.
- **NEW SYMBOL** c_k, d_k and Φ: probit transition coefficients and the normal cdf.

Likely needed in Chapters 3–5:

- **NEW SYMBOL** $f_{k,t} = f(r_t \mid S_t = k, \mathcal{F}_{t-1})$: regime density shorthand.
- **NEW SYMBOL** $\xi_{k,t|t-1}$: already proposed; used heavily here.
- **NEW SYMBOL** $w_{jk,t} = P(S_{t-1} = j \mid \mathcal{F}_{t-1}, S_t = k)$: Klaassen's backward weight.
- **NEW SYMBOL** $\sigma^2_{k,\tau|t-1}$: regime-k variance forecast for date τ.
- **NEW SYMBOL** $\bar{\sigma}_k^2$: unconditional regime variance (extends the approved $\bar{\sigma}^2$).
- **NEW SYMBOL** $\mathbf{A}$: Klaassen's existence matrix (bold, to avoid any clash).
- **NEW SYMBOL** $r_{t,h}$, $v_{t,h}$: h-day return and realised volatility.
- **NEW SYMBOL** κ_0, κ_1: forecast-efficiency coefficients.
- **NEW SYMBOL** $\tilde{R}^2$: Klaassen's restricted R^2.

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